

08 Geometric Transformations

8.4 Shearing

Horizontal and vertical shear matrices, shear angle; same-axis vs. different-axis composition; scale-shear decomposition; six basic 3D shears



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

Shear keeps one direction's coordinate fixed while shifting the other in proportion to it

2

Use the shear angle ϑ to describe direction and strength: $\cot(\vartheta)$ unifies horizontal and vertical shear

3

A shear matrix's determinant is always 1 — area (or volume) is preserved

4

Shear matrices are invertible; the inverse is still an upper or lower triangular matrix of the same type

5

Two shears along the same axis merge into a single shear, and the order can be swapped

6

Two shears along different axes do not commute — the order changes the result

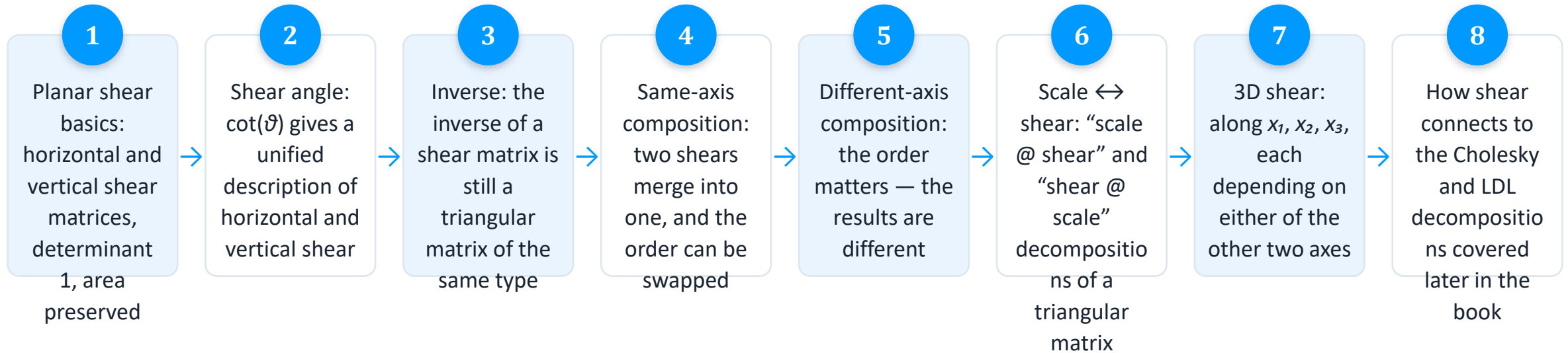
7

A diagonal matrix with nonzero entries can be written as “scale @ shear” or “shear @ scale”

8

In 3D, a shear along one axis can depend on either of the other two axes — six basic shears in all

From Planar Shear to Six Basic 3D Shears



Seven Lenses for Every Transform in This Chapter

- 1 Matrix-multiplication view: how the matrix acts on $\mathbf{0}$, $\mathbf{e}_1$, $\mathbf{e}_2$, and the all-ones vector $\mathbf{1}$
- 2 The determinant: $|\det|$ greater than 1? Positive, negative, or zero? Determinants of special matrices
- 3 How the determinant reflects the area or volume change caused by the transform
- 4 The inverse (does it exist?) and the inverse transform; inverses of special matrices
- 5 Linear-combination view: parallelogram/parallelepiped grids, and whether dimension drops
- 6 Does the transform matrix form a basis, and what kind of basis?
- 7 Properties of the transform matrix: symmetric, diagonal, orthogonal, invertible, commutative

Horizontal shear tilts a figure sideways, leaving x_2 unchanged

$$[z_1; z_2] = [1, k; 0, 1] @ [x_1; x_2]$$

- A point (x_1, x_2) moves to $(x_1 + k \cdot x_2, x_2)$ — the horizontal coordinate shifts, the vertical one stays fixed
- The matrix is upper triangular with determinant 1, so this transform never changes area
- k is not a translation but a ratio: the size of x_2 sets the offset, its sign sets the direction

Figure 1 · Two Examples of Horizontal Shear

- (a) $k > 0$ tilts right; (b) $k < 0$ tilts left — the larger $|k|$ is, the stronger the shear

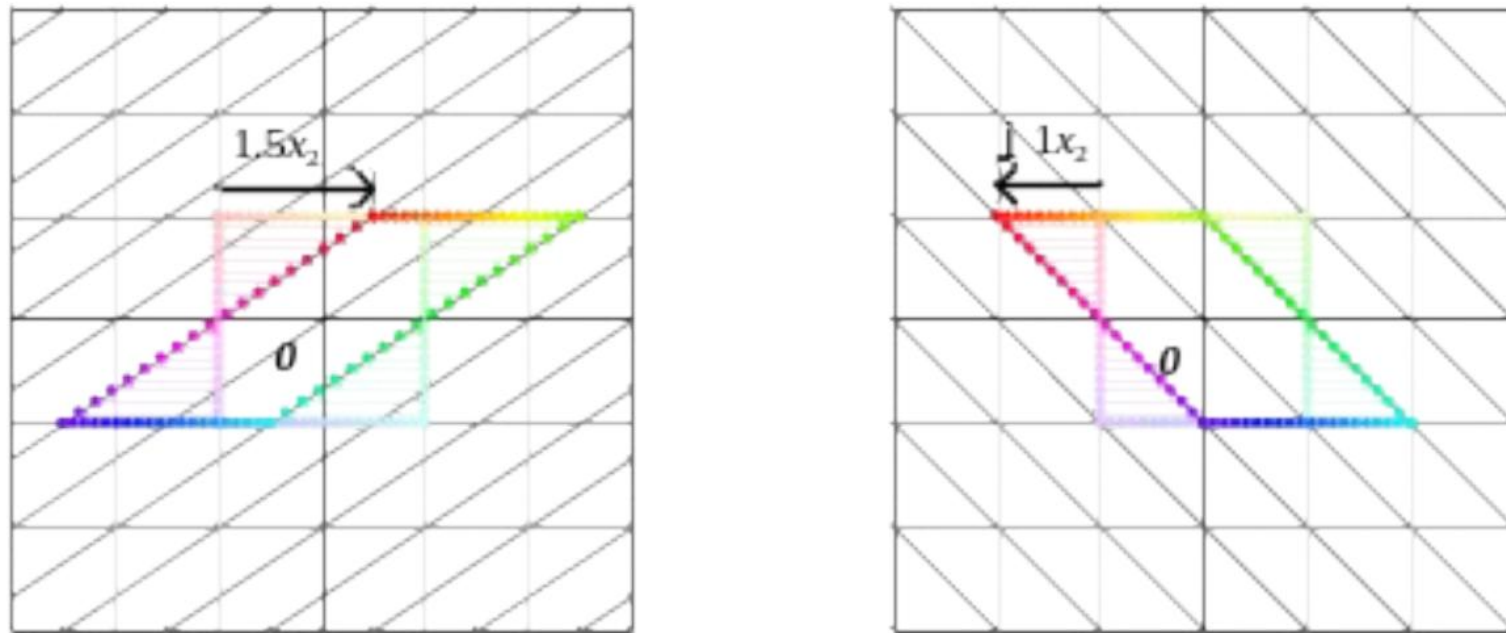


图 1. 两个水平剪切变换的例子

Vertical shear tilts a figure up and down, leaving x_1 unchanged

$$[z_1; z_2] = [1, 0; k, 1] @ [x_1; x_2]$$

- A point (x_1, x_2) moves to $(x_1, x_2 + k \cdot x_1)$ — the horizontal coordinate stays fixed, the vertical one shifts
- The matrix is lower triangular, again with determinant 1 — area is preserved

Figure 2 · Two Examples of Vertical Shear

- (a)(b) two shears with different k — $k > 0$ shifts up, $k < 0$ shifts down; larger $|k|$ means a bigger shift

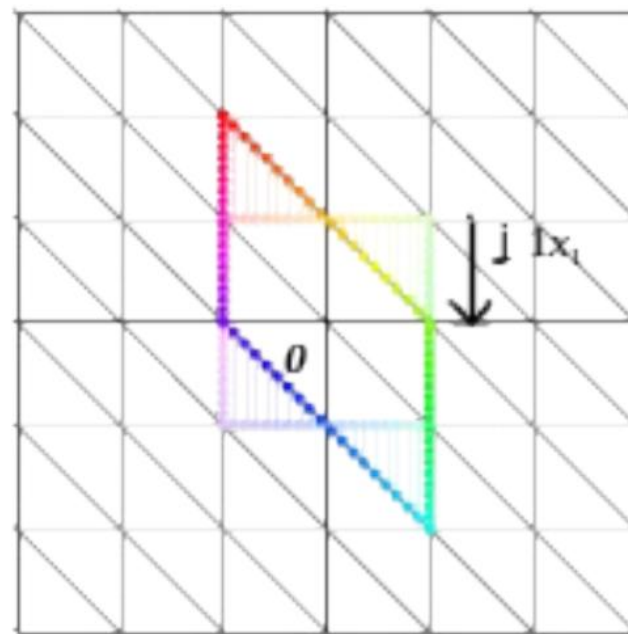
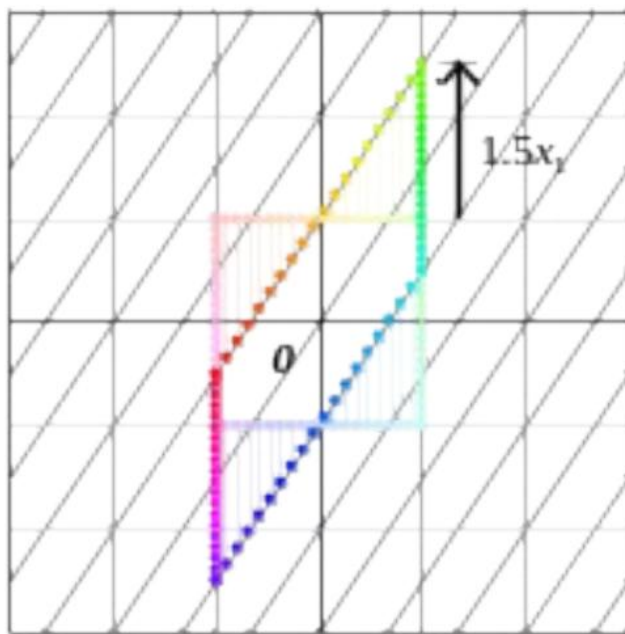


图 2. 两个竖直剪切变换的例子

Describing horizontal shear with a shear angle θ ; $\cot(\theta)$ is its cotangent

$$[z_1; z_2] = [1, \cot(\theta); 0, 1] @ [x_1; x_2]$$

- A point (x_1, x_2) moves to $(x_1 + \cot(\vartheta) \cdot x_2, x_2)$ — the horizontal coordinate shifts, the vertical one stays fixed
- When $0^\circ < \vartheta < 90^\circ$, the figure tilts right; when $0^\circ > \vartheta > -90^\circ$, it tilts left

Figure 3 · Two Examples of Horizontal Shear Using the Shear Angle

- (a) $\vartheta=60^\circ$, tilting right; (b) $\vartheta=-30^\circ$, tilting left

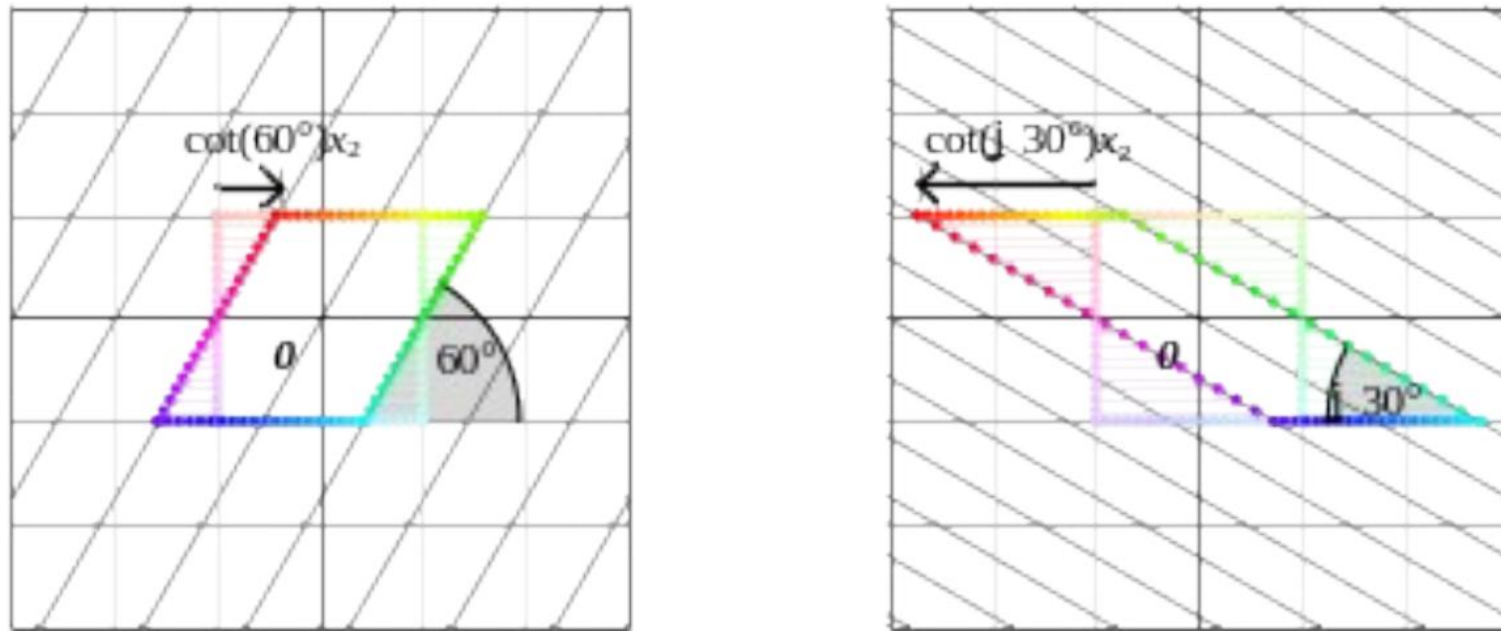


图 3. 两个水平剪切变换的例子，利用剪切角

Describing vertical shear with a shear angle θ

$$[z_1; z_2] = [1, 0; \cot(\theta), 1] @ [x_1; x_2]$$

- A point (x_1, x_2) moves to $(x_1, x_2 + \cot(\vartheta) \cdot x_1)$ — the horizontal coordinate stays fixed, the vertical one shifts
- When $0^\circ < \vartheta < 90^\circ$, the figure tilts up; when $0^\circ > \vartheta > -90^\circ$, it tilts down

Figure 4 · Two Examples of Vertical Shear Using the Shear Angle

- (a) $\vartheta=60^\circ$, tilting up; (b) $\vartheta=-30^\circ$, tilting down

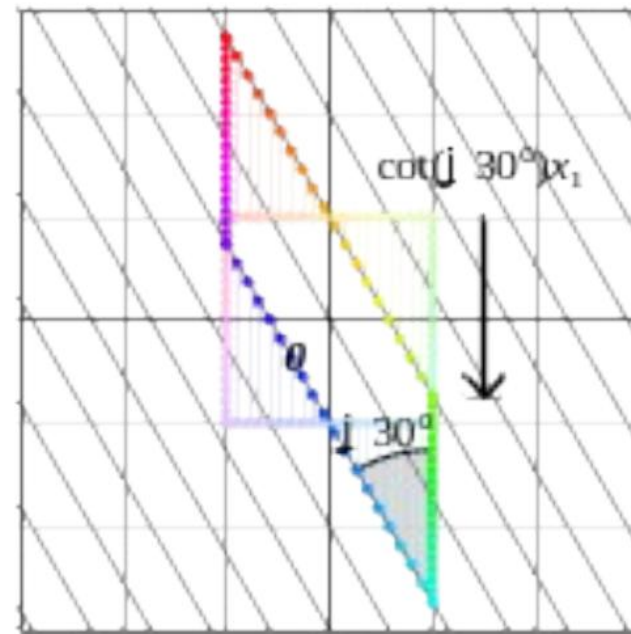
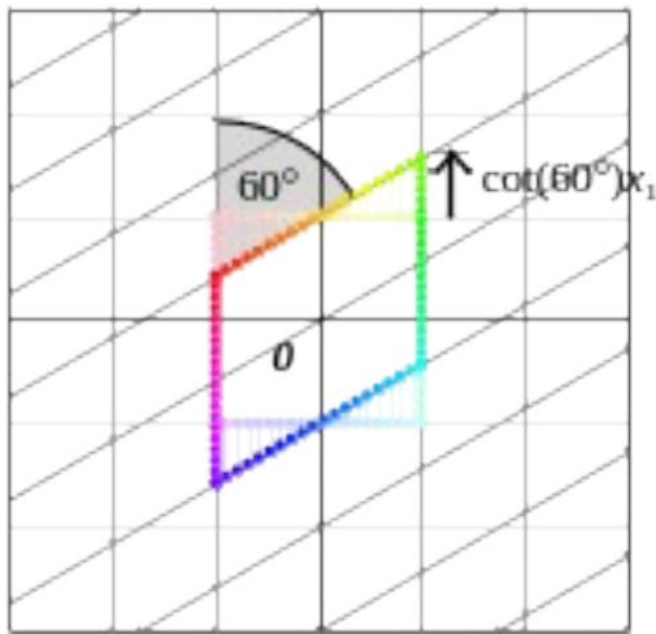


图 4. 两个竖直剪切变换的例子，利用剪切角

Inverse of shear along x_1 — the inverse of an upper triangular matrix stays upper triangular

$$[1, k; 0, 1]^{-1} = [1, -k; 0, 1]$$

- The inverse shear matrix is still a horizontal shear, only with shear factor $-k$ instead of k
- Multiplying the two together gives the identity I , confirming the inverse is correct

Figure 5 · Shear Along the Horizontal Axis, and Its Inverse

- Shearing by k , then by $-k$, restores the original figure

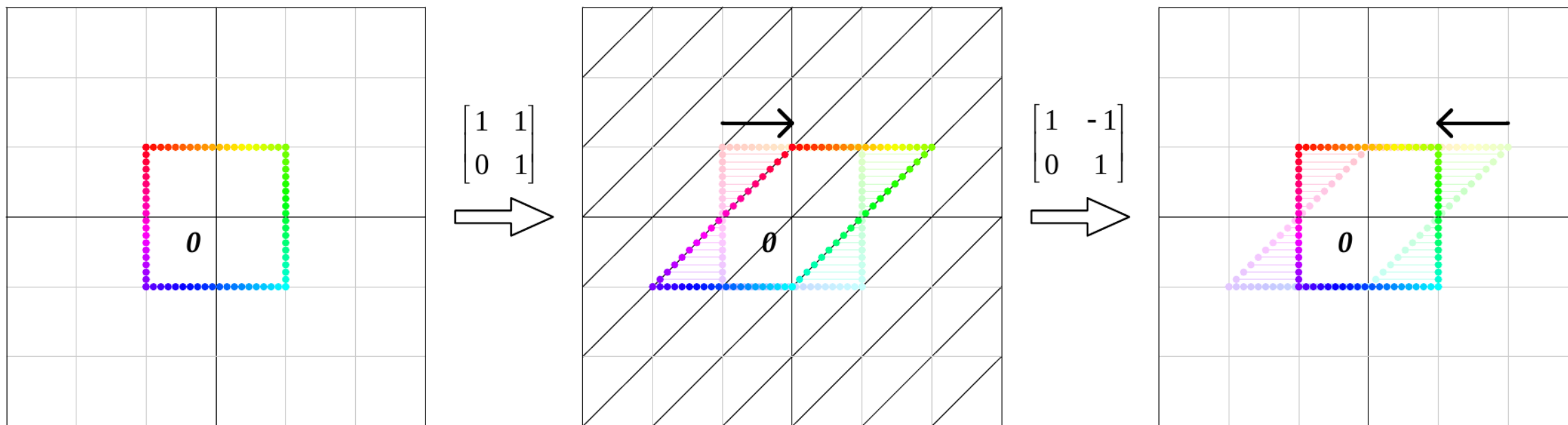


图 5. 沿横轴剪切，以及逆操作

Inverse of shear along x_2 — the inverse of a lower triangular matrix stays lower triangular

$$[1, 0; k, 1]^{-1} = [1, 0; -k, 1]$$

- The inverse of vertical shear likewise just negates the shear factor k
- This holds for both upper and lower triangular matrices — an important property of triangular matrices

Figure 6 · Shear Along the Vertical Axis, and Its Inverse

- Shearing by k , then by $-k$, restores the original figure

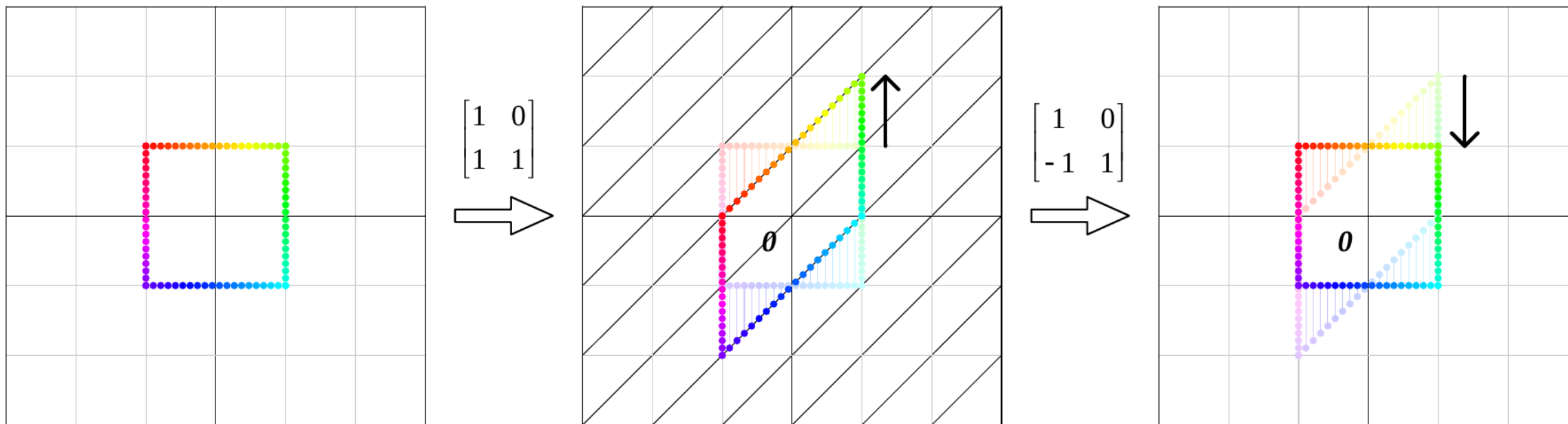


图 6. 沿纵轴剪切，以及逆操作

Two shears along the same axis merge into one, and the order can be swapped

$$[1, k_1; 0, 1] @ [1, k_2; 0, 1] = [1, k_2; 0, 1] @ [1, k_1; 0, 1] = [1, k_1 + k_2; 0, 1]$$

- This holds for two consecutive shears along either x_1 or x_2 — the result depends only on the sum of the two factors
- This is one special case where matrix multiplication does commute

Figure 7 · Composing Two Shears Along the Horizontal Axis

- Shearing by k_1 then k_2 , or reversing the order, gives exactly the same figure

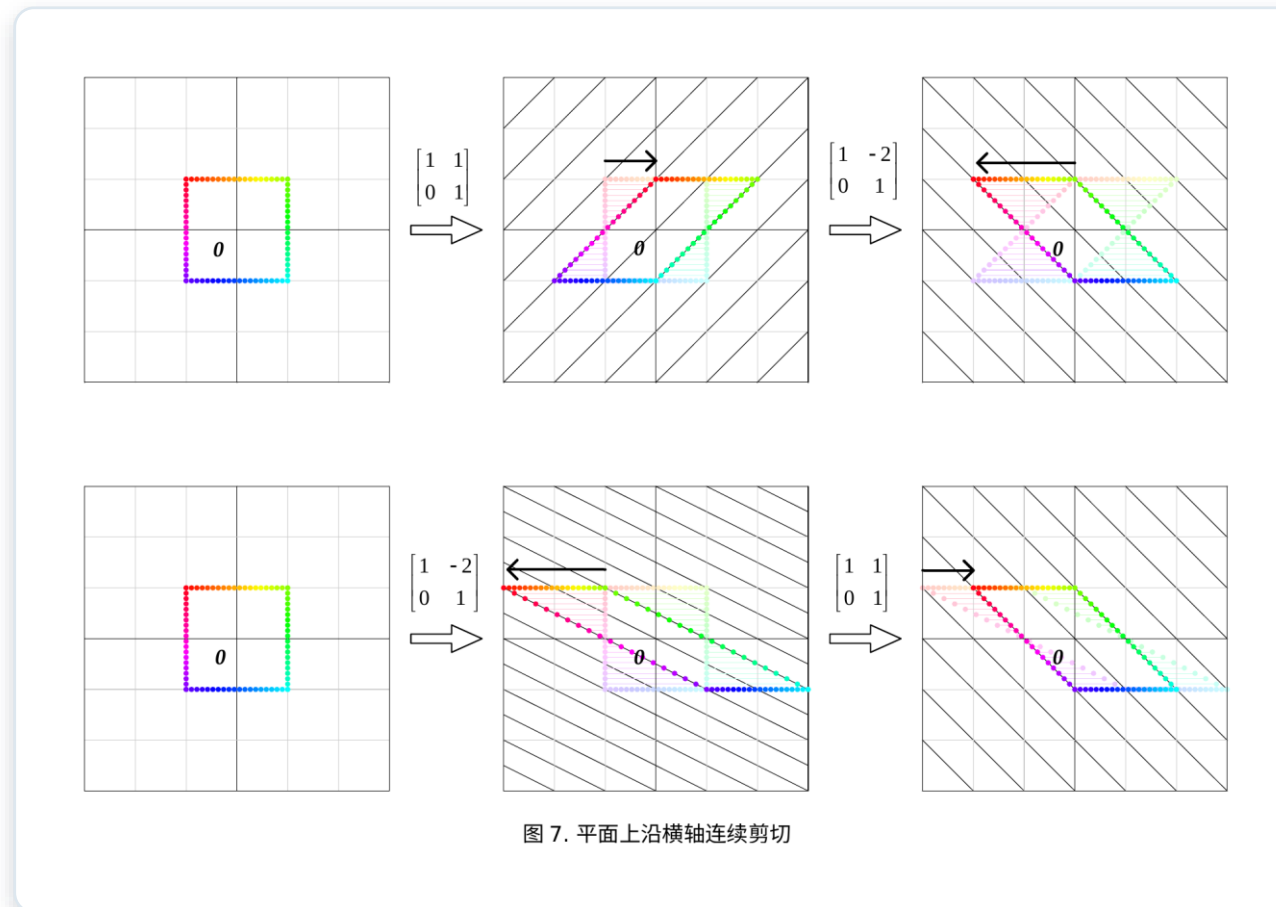
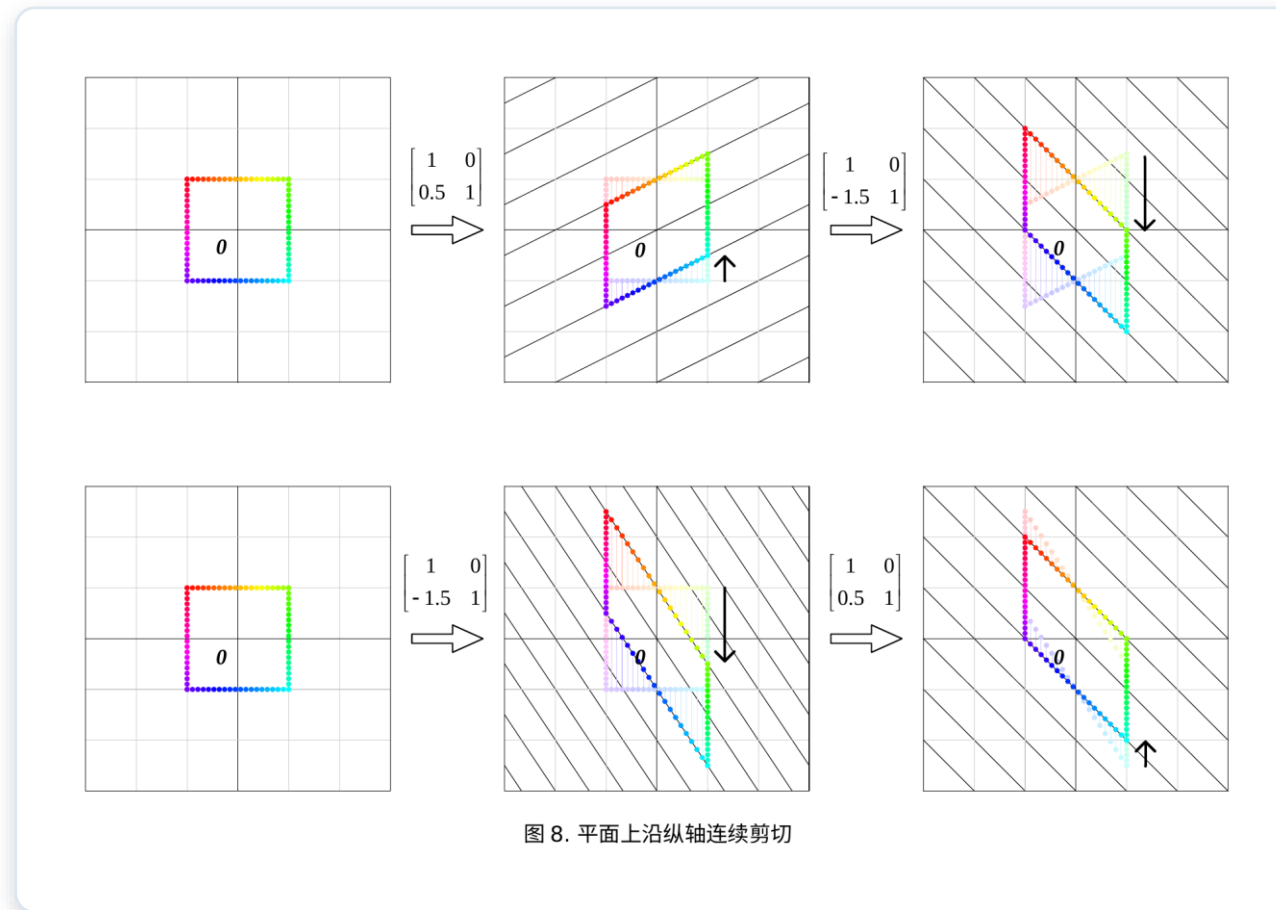


Figure 8 · Composing Two Shears Along the Vertical Axis

- The vertical case is the same: reversing the order of the two shears leaves the result unchanged



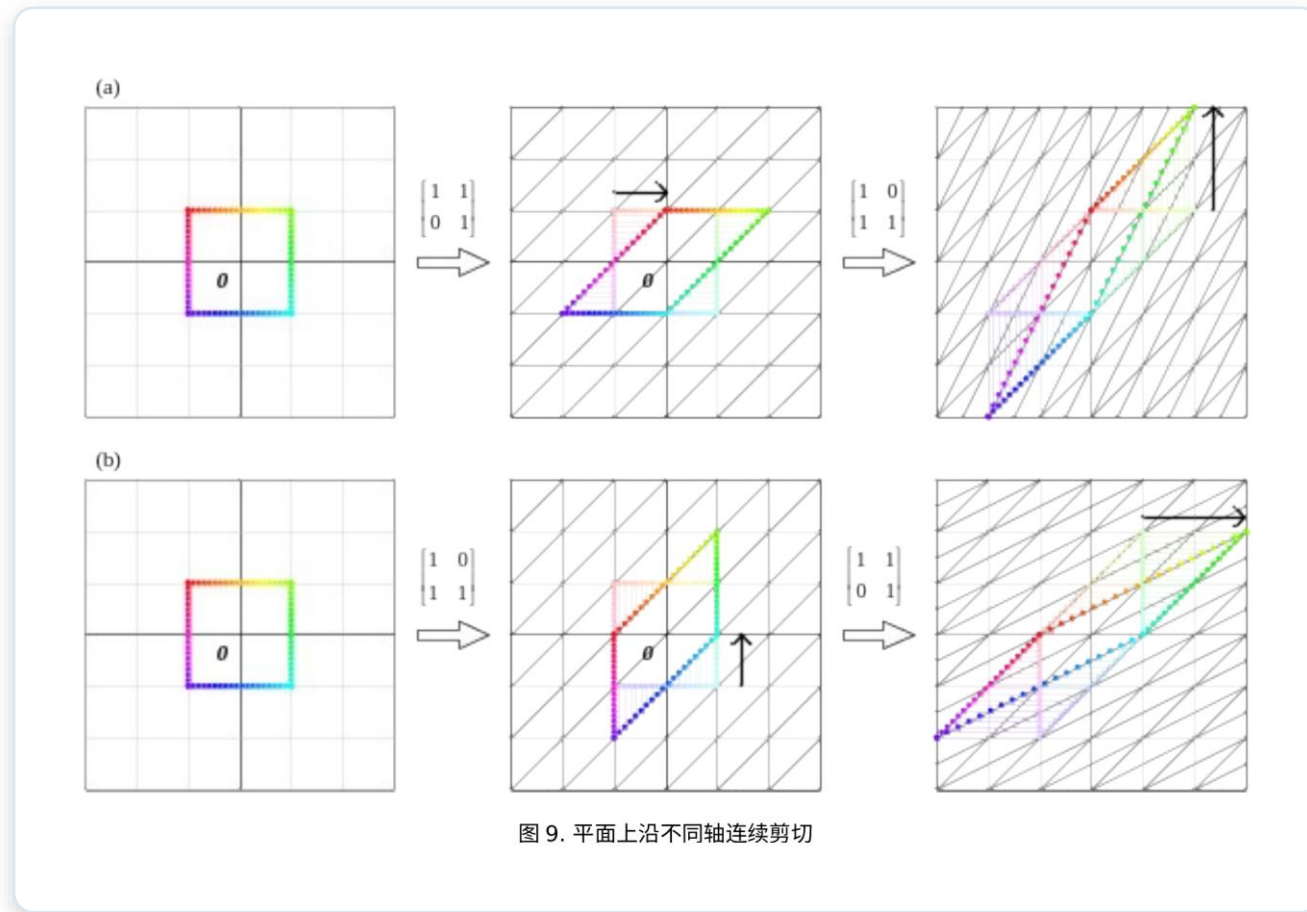
Shear along x_1 then x_2 — and the reverse order — give completely different results

$$[1,0; k_2,1] @ [1,k_1; 0,1] \neq [1,k_1; 0,1] @ [1,0; k_2,1]$$

- (a) x_1 then x_2 : result is $[1,k_1; k_2,k_1 \cdot k_2 + 1]$
- (b) x_2 then x_1 : result is $[k_1 \cdot k_2 + 1, k_1; k_2, 1]$ — completely different from (a)

Figure 9 · Composing Shears Along Different Axes in the Plane

- (a) shear along x_1 then x_2 ; (b) shear along x_2 then x_1 — different results



SCALE → SHEAR

Any upper triangular 2×2 matrix with nonzero diagonal can be written “ $S(\text{scale})@K(\text{horizontal shear})$ ”

$$[2/3, 1/3; 0, 3/2] = [2/3, 0; 0, 3/2] @ [1, 1/2; 0, 1]$$

- SKx acts on a column vector x by first applying K (horizontal shear), then S (scale)
- This decomposition reveals the link between triangular matrices and the two basic transforms scale and shear

Figure 10 · Horizontal Shear (K) $\rightarrow$ Scale (S)

- First sheared along the horizontal axis into a parallelogram, then scaled to the target shape

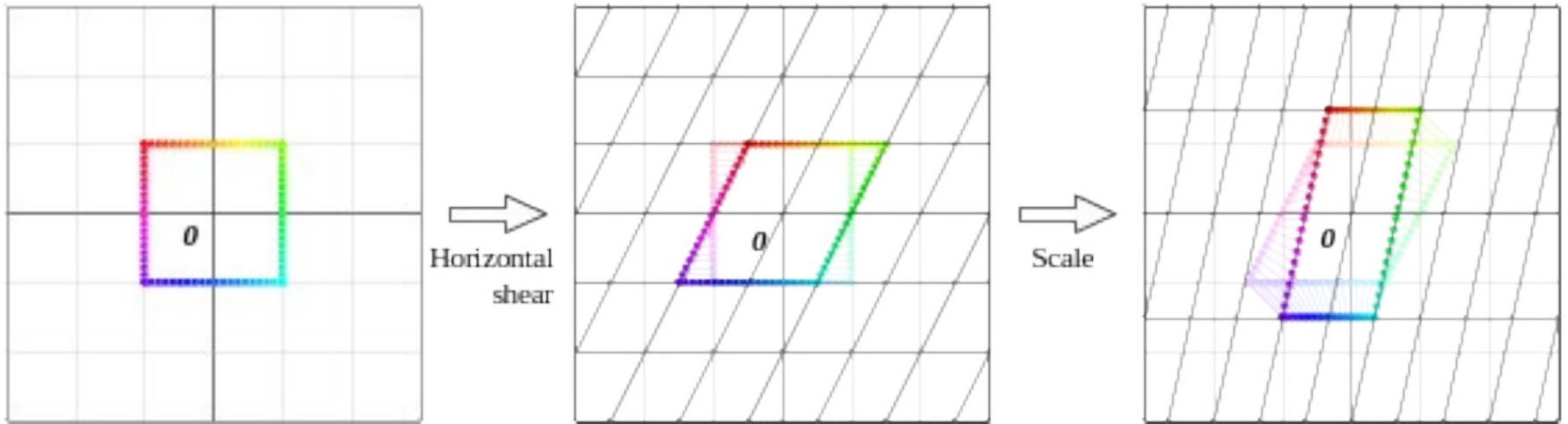


图 10. 先横轴剪切 (K) $\rightarrow$ 后缩放 (S)

SCALE → SHEAR

Similarly, a lower triangular 2×2 matrix can be written
“ $S(\text{scale}) @ K(\text{vertical shear})$ ”

$$[2/3, 0; 1/3, 3/2] = [2/3, 0; 0, 3/2] @ [1, 0; 2/9, 1]$$

- SKx acts on a column vector x by first applying K (vertical shear), then S (scale)
- This will help us understand the Cholesky and LDL decompositions covered later in the book

Figure 11 · Vertical Shear (K) $\rightarrow$ Scale (S)

- First sheared along the vertical axis into a parallelogram, then scaled to the target shape

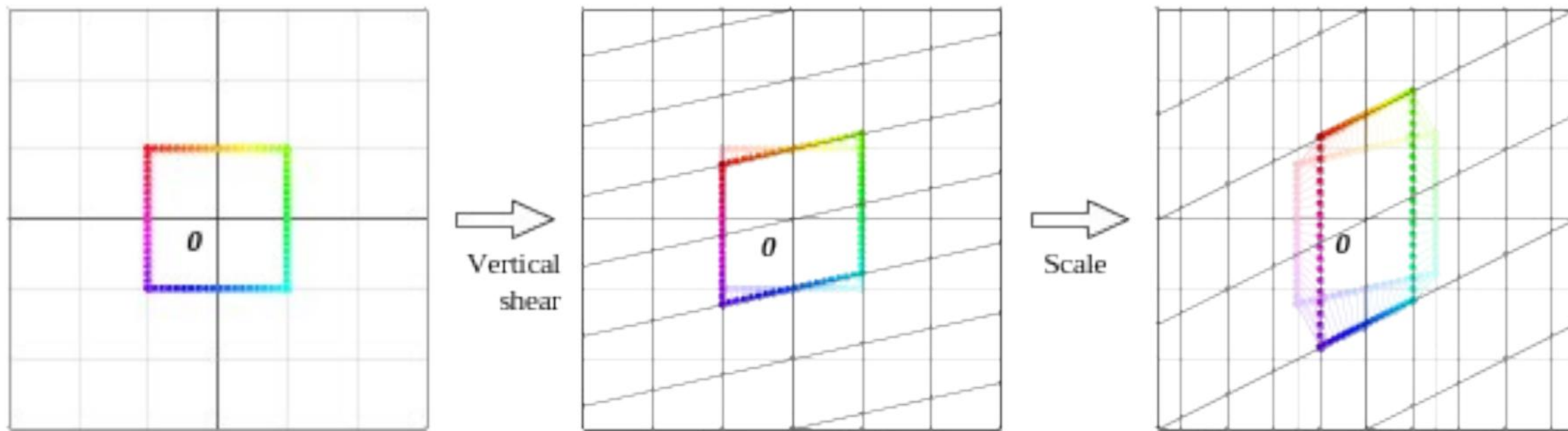


图 11. 先纵轴剪切 (K) $\rightarrow$ 后缩放 (S)

SHEAR → SCALE

Conversely, the same upper triangular matrix can be written
“ $K(\text{horizontal shear})@S(\text{scale})$ ”

$$[2/3, 1/3; 0, 3/2] = [1, 2/9; 0, 1] @ [2/3, 0; 0, 3/2]$$

- KSx acts on a column vector x by first applying S (scale), then K (horizontal shear)
- Figures 10 and 12 give the same final result; the products SK and KS are transposes of each other

Figure 12 · Scale (S) → Horizontal Shear (K)

- A square is first scaled, then sheared along the horizontal axis — the final shape matches Figure 10

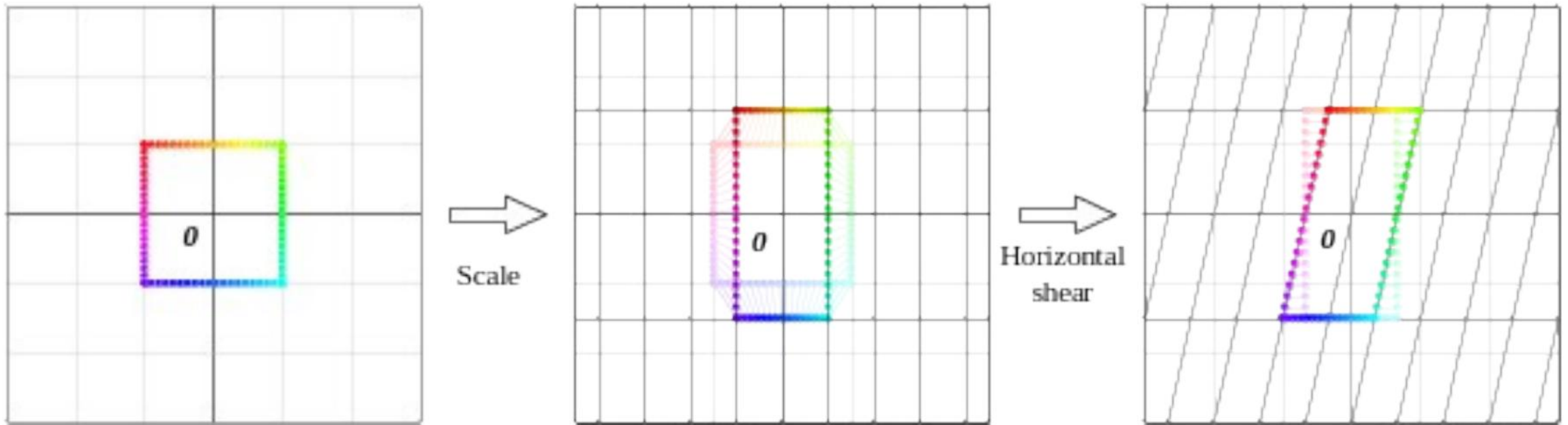


图 12. 先缩放 (S) → 后横轴剪切 (K)

SHEAR → SCALE

Similarly, a lower triangular matrix can be written “ $K(\text{vertical shear})@S(\text{scale})$ ”

$$[2/3, 0; 1/3, 3/2] = [1, 0; 1/2, 1] @ [2/3, 0; 0, 3/2]$$

- KSx acts on a column vector x by first applying S (scale), then K (vertical shear)
- Figure 13 matches Figure 11; again, the products SK and KS are transposes of each other

Figure 13 · Scale (S) → Vertical Shear (K)

- A square is first scaled, then sheared along the vertical axis — the final shape matches Figure 11

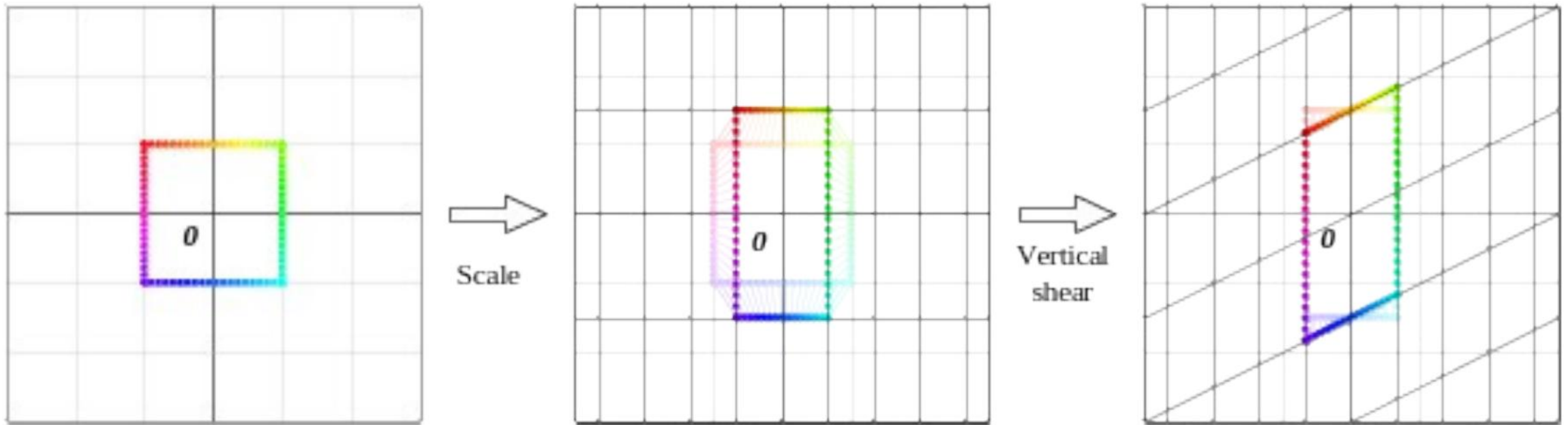


图 13. 先缩放 (S) → 后纵轴剪切 (K)

3D SHEAR ALONG X1

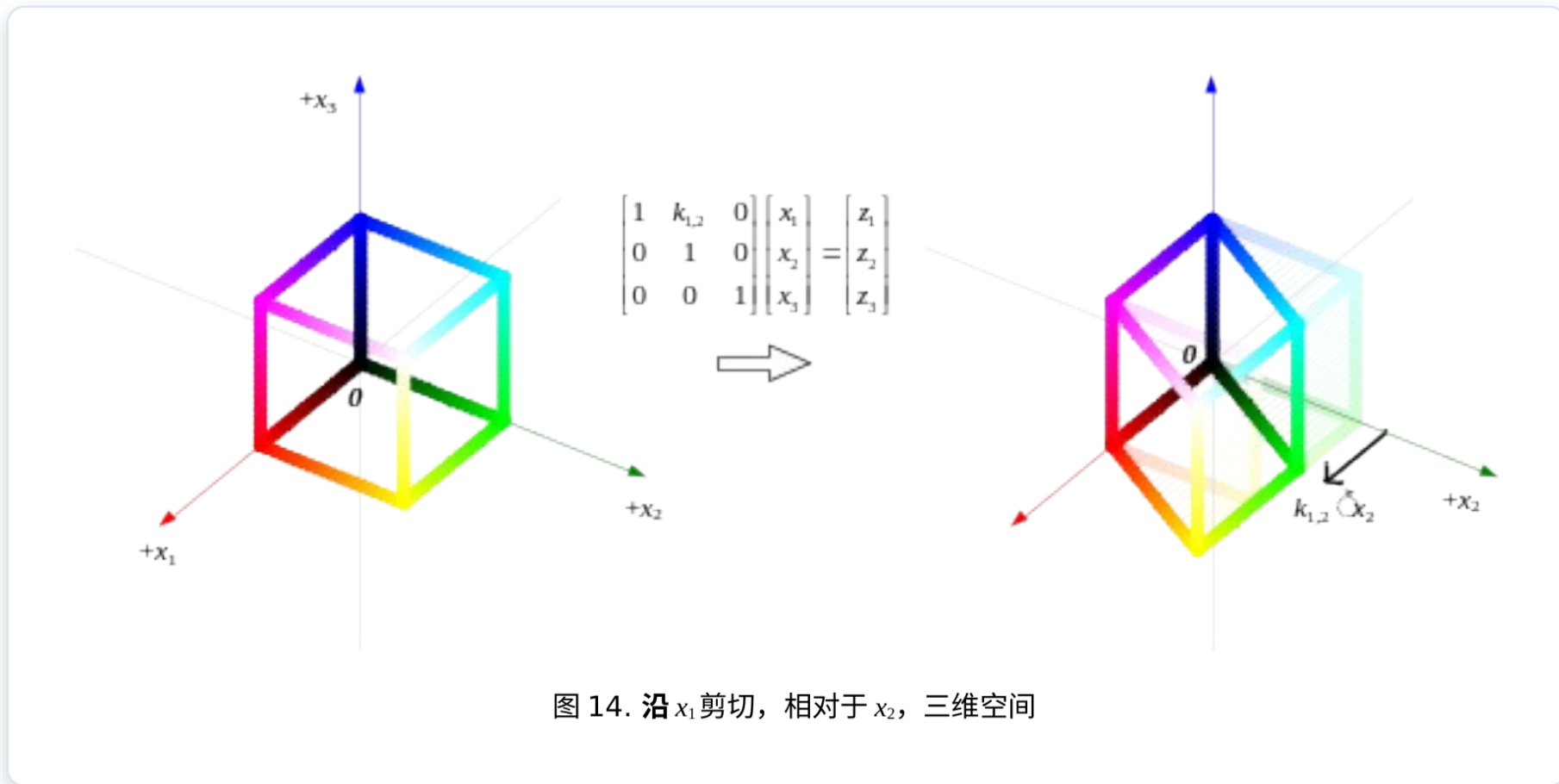
The shear in x_1 varies with x_2 ; determinant 1 means volume is preserved

$$[z_1; z_2; z_3] = [1, k_{1,2}, 0; 0, 1, 0; 0, 0, 1] @ [x_1; x_2; x_3]$$

- Expanded: $z_1 = x_1 + k_{1,2} \cdot x_2$, $z_2 = x_2$, $z_3 = x_3$
- 3D shear can be viewed along x_1 , x_2 , x_3 , and each axis can in turn depend on either of the other two

Figure 14 · Shear Along x_1 , Relative to x_2 , in 3D

- The cube tilts along x_1 , with the amount set by the x_2 coordinate; volume is preserved



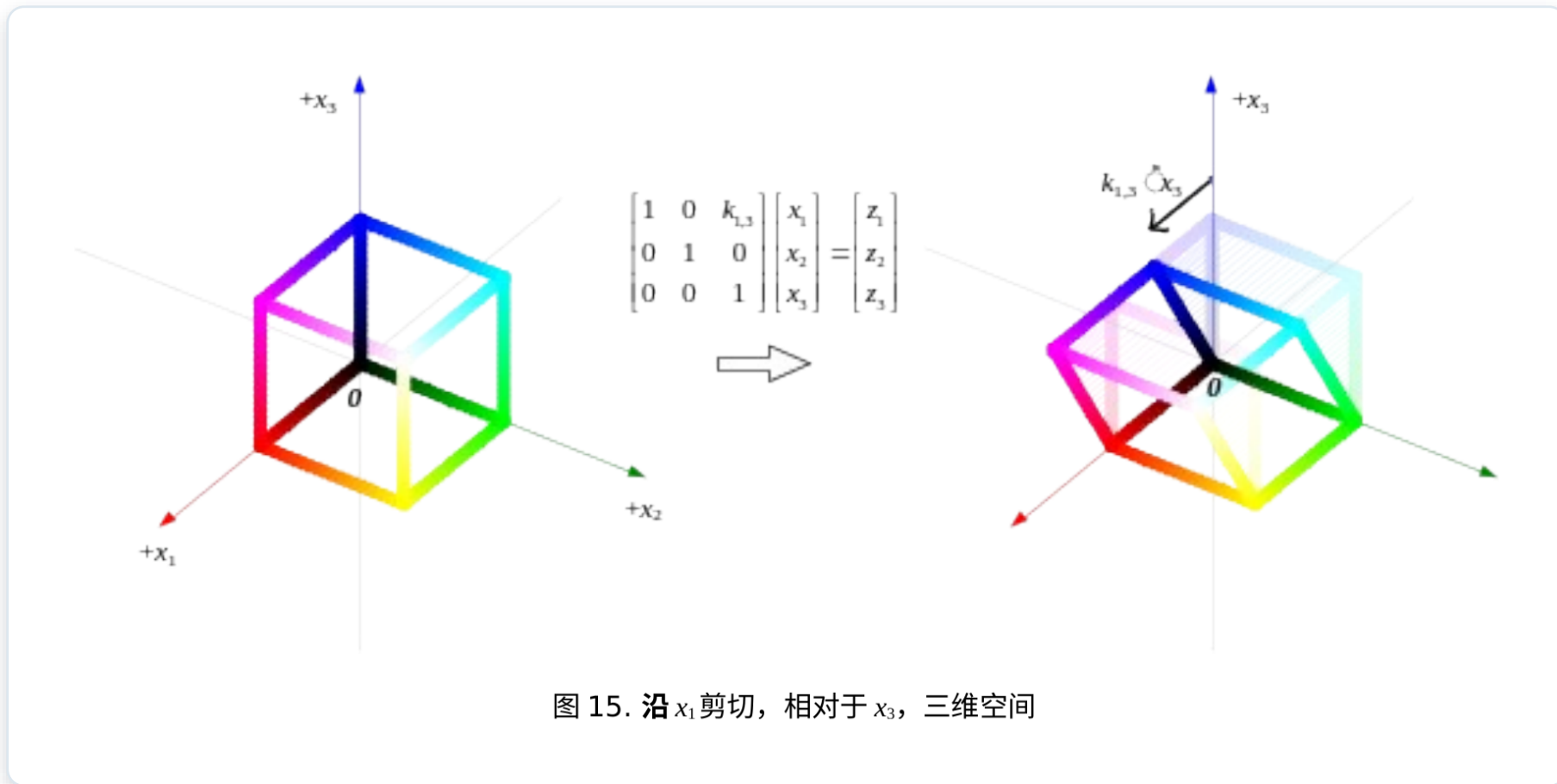
The shear in x_1 varies with x_3 — again a volume-preserving transform with determinant 1

$$[z_1; z_2; z_3] = [1, 0, k_{1,3}; 0, 1, 0; 0, 0, 1] @ [x_1; x_2; x_3]$$

- Expanded: $z_1 = x_1 + k_{1,3} \cdot x_3$, $z_2 = x_2$, $z_3 = x_3$
- These two matrix products commute — two shears along the same axis satisfy the commutative law

Figure 15 · Shear Along x_1 , Relative to x_3 , in 3D

- The cube tilts along x_1 , with the amount set by the x_3 coordinate; volume is preserved



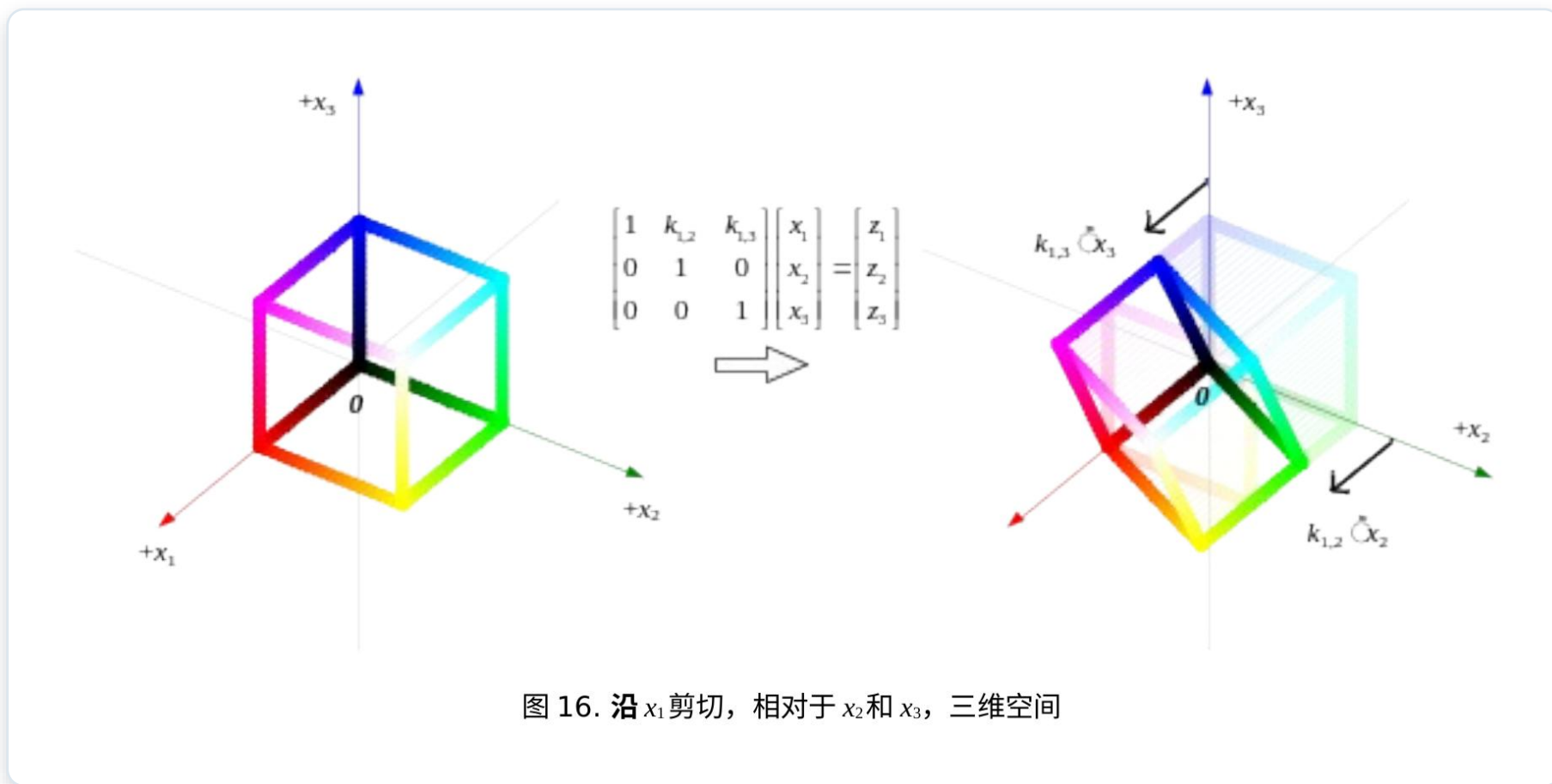
Merging the two shears relative to x_2 and x_3 into a single shear

$$[z_1; z_2; z_3] = [1, k_{1,2}, k_{1,3}; 0, 1, 0; 0, 0, 1] @ [x_1; x_2; x_3]$$

- Expanded: $z_1 = x_1 + k_{1,2} \cdot x_2 + k_{1,3} \cdot x_3$, $z_2 = x_2$, $z_3 = x_3$
- This confirms the two shears along x_1 give a commuting matrix product, satisfying the commutative law

Figure 16 · Shear Along x_1 , Relative to x_2 and x_3 , in 3D

- The cube tilts along x_1 , with the amount set jointly by the x_2 and x_3 coordinates



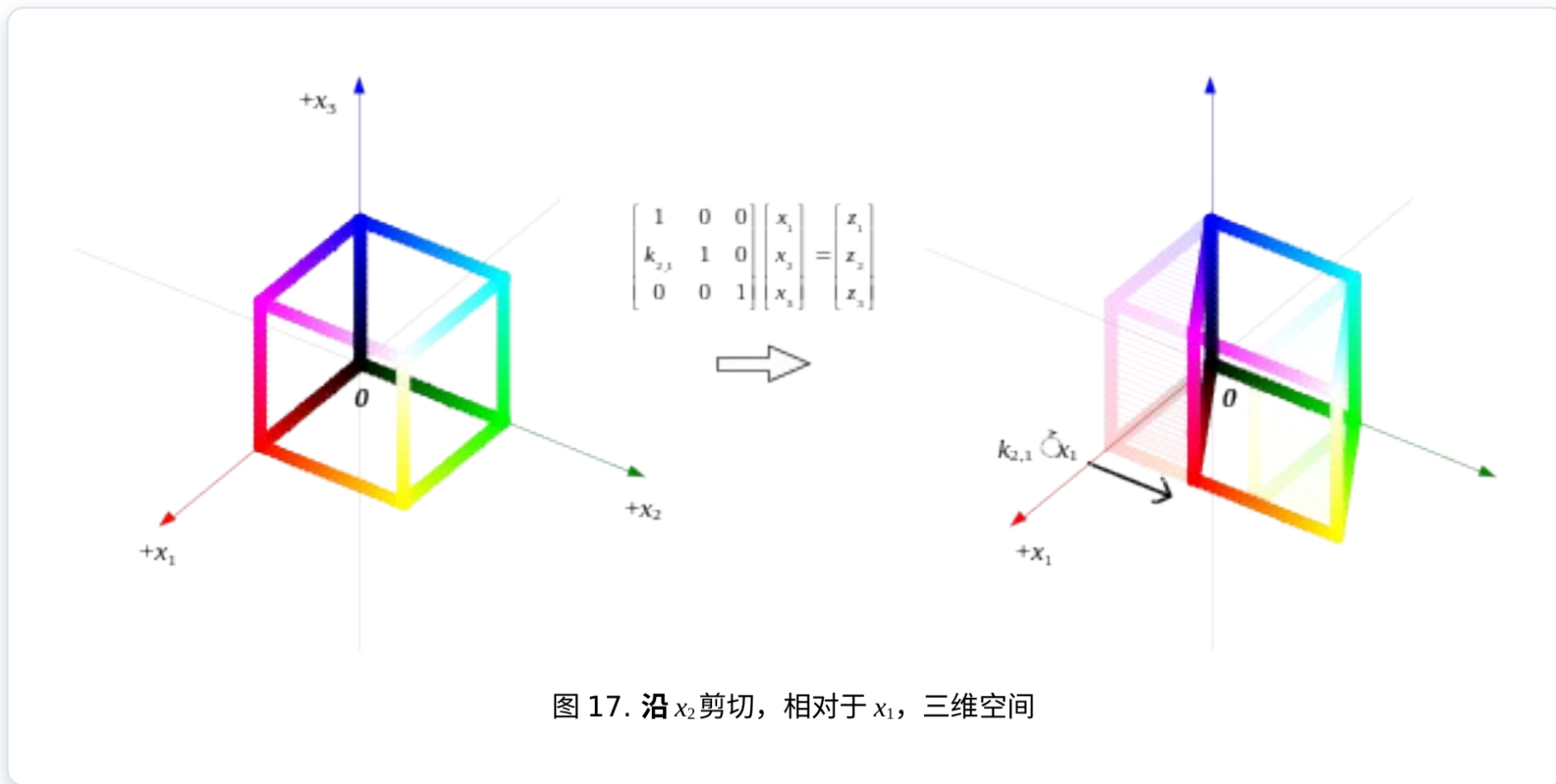
The shear in x_2 varies with x_1

$$[z_1; z_2; z_3] = [1, 0, 0; k_{2,1}, 1, 0; 0, 0, 1] @ [x_1; x_2; x_3]$$

- Expanded: $z_1 = x_1$, $z_2 = k_{2,1} \cdot x_1 + x_2$, $z_3 = x_3$
- Shear along x_2 also has two reference directions, x_1 and x_3

Figure 17 · Shear Along x_2 , Relative to x_1 , in 3D

- The cube tilts along x_2 , with the amount set by the x_1 coordinate; volume is preserved



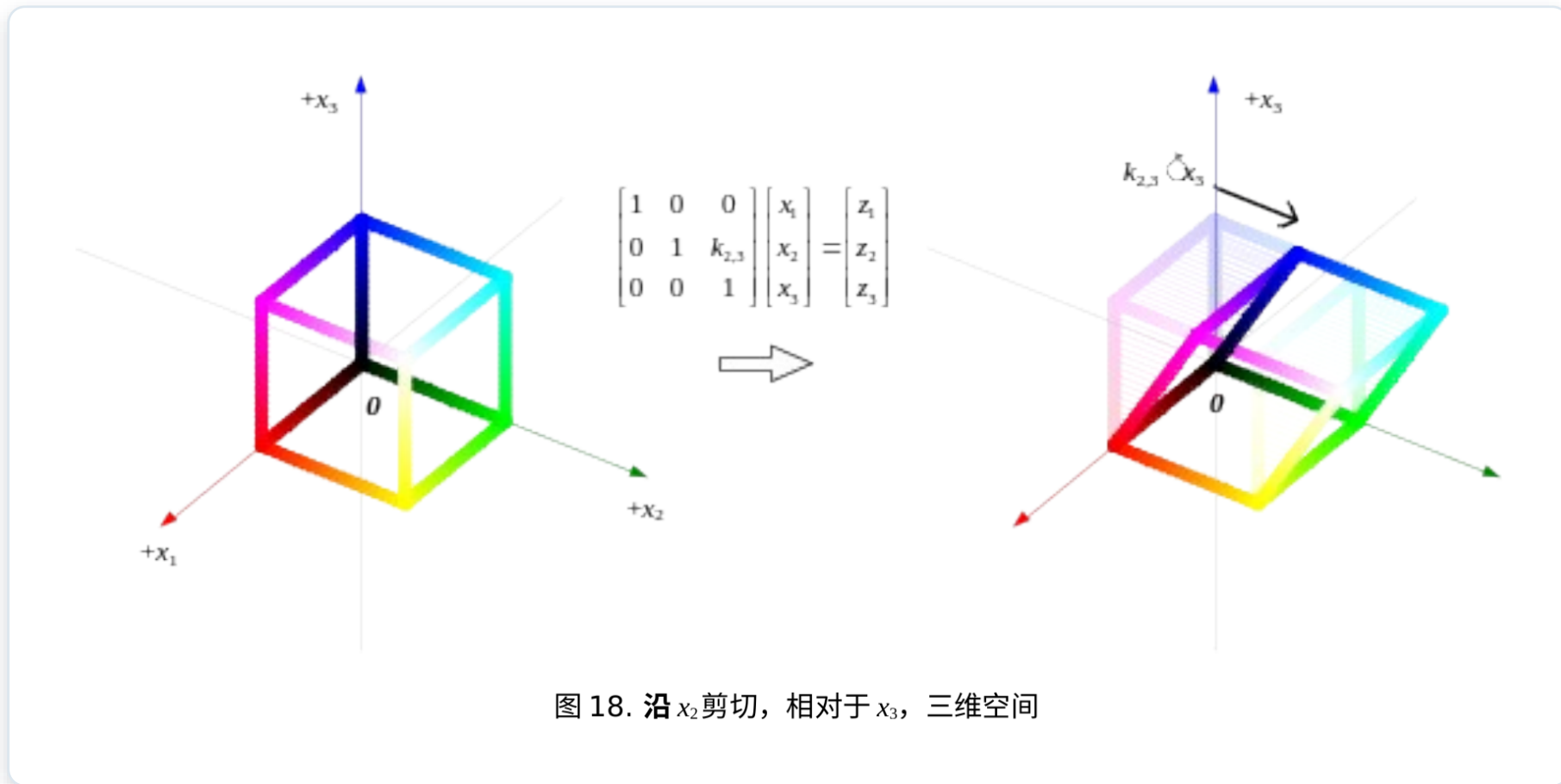
The shear in x_2 varies with x_3

$$[z_1; z_2; z_3] = [1, 0, 0; 0, 1, k_{2,3}; 0, 0, 1] @ [x_1; x_2; x_3]$$

- Expanded: $z_1=x_1, z_2=x_2+k_{2,3} \cdot x_3, z_3=x_3$
- Combining the shears relative to x_1 and x_3 gives the merged Figure 19

Figure 18 · Shear Along x_2 , Relative to x_3 , in 3D

- The cube tilts along x_2 , with the amount set by the x_3 coordinate; volume is preserved



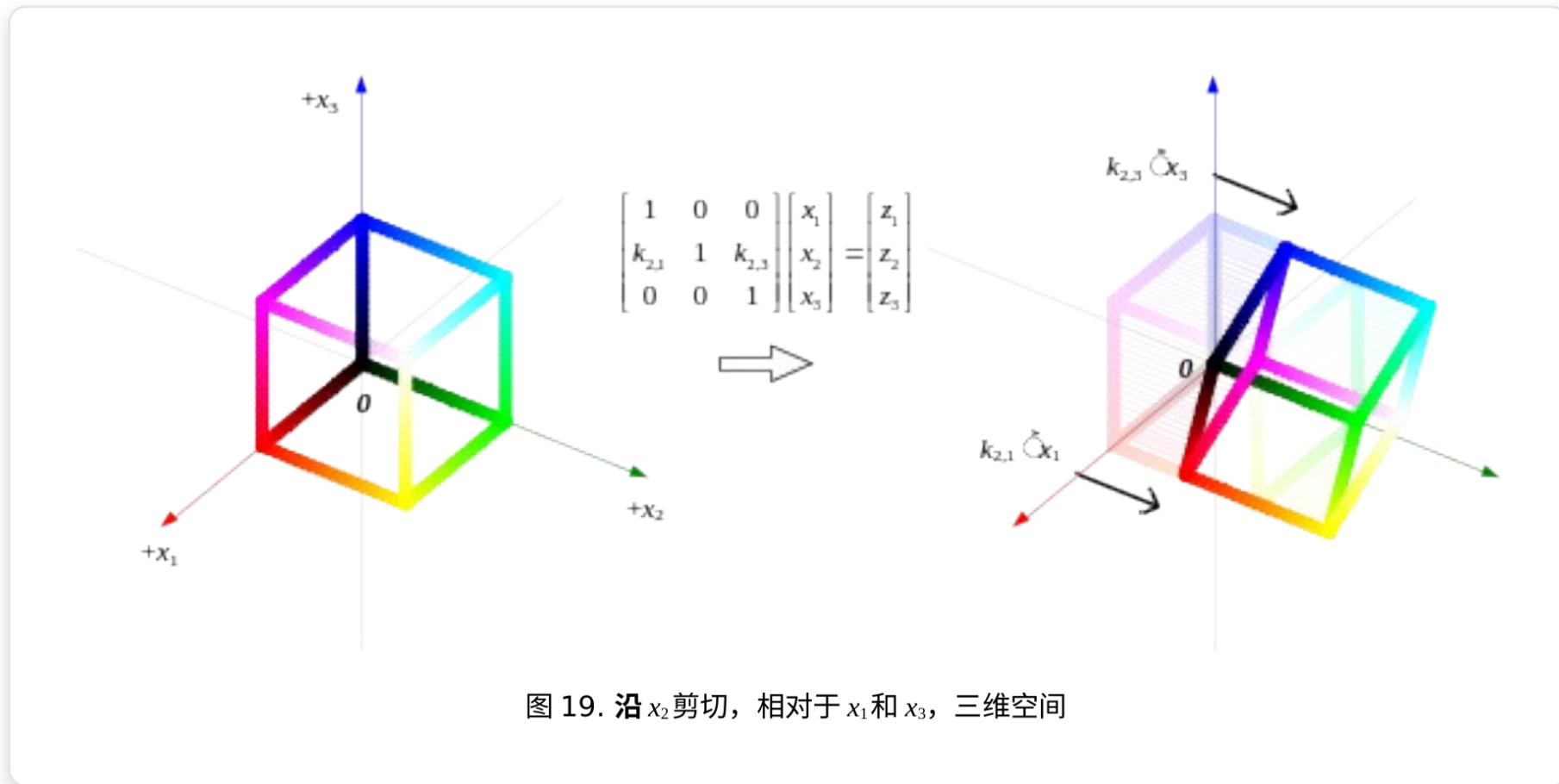
Merging the two shears relative to x_1 and x_3 into a single shear

$$[z_1; z_2; z_3] = [1, 0, 0; k_{2,1}, 1, k_{2,3}; 0, 0, 1] @ [x_1; x_2; x_3]$$

- Expanded: $z_1 = x_1$, $z_2 = k_{2,1} \cdot x_1 + x_2 + k_{2,3} \cdot x_3$, $z_3 = x_3$
- The two shears along x_2 likewise satisfy the commutative law for matrix multiplication

Figure 19 · Shear Along x_2 , Relative to x_1 and x_3 , in 3D

- The cube tilts along x_2 , with the amount set jointly by the x_1 and x_3 coordinates



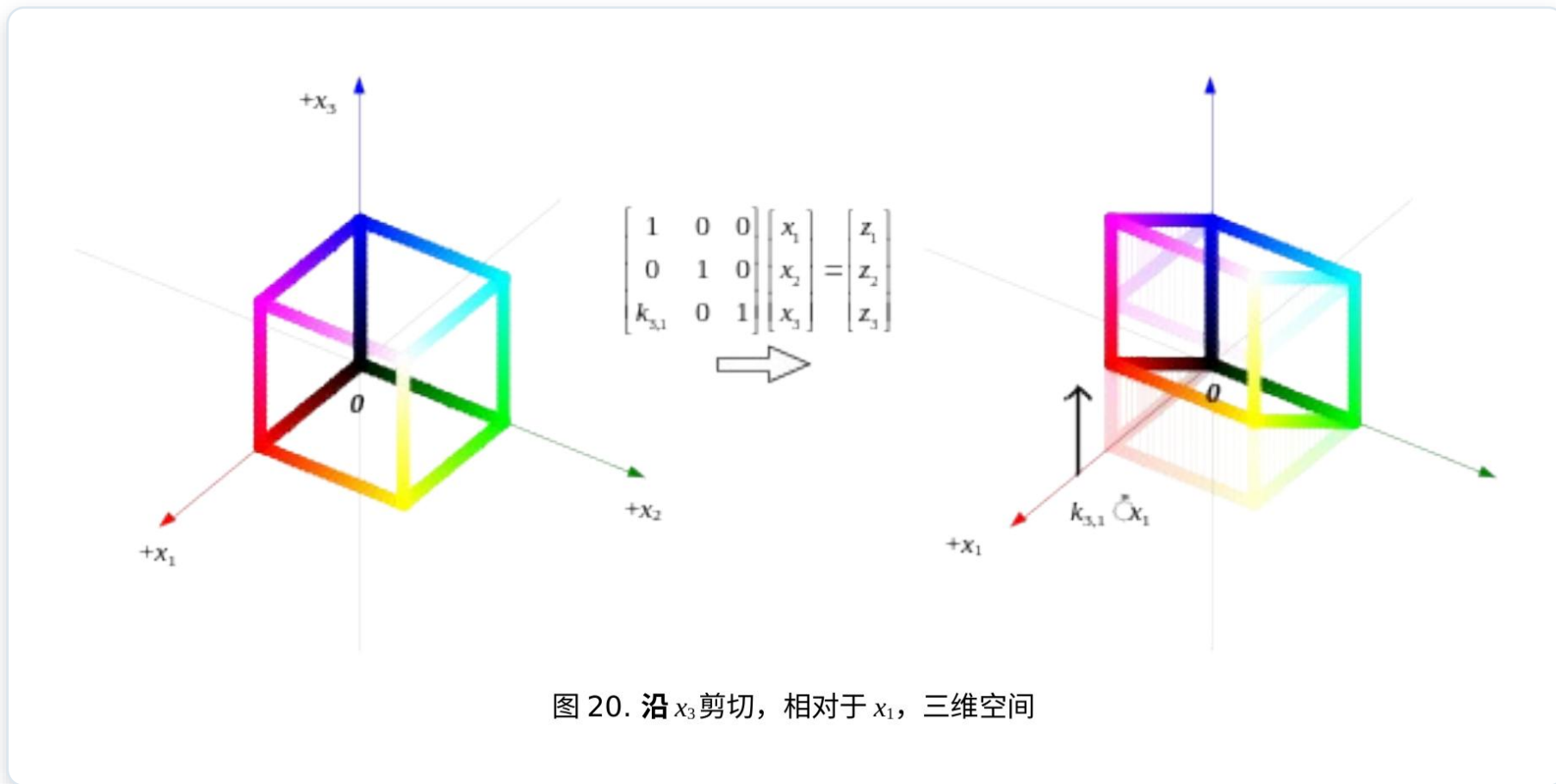
The shear in x_3 varies with x_1

$$[z_1; z_2; z_3] = [1, 0, 0; 0, 1, 0; k_{3,1}, 0, 1] @ [x_1; x_2; x_3]$$

- Expanded: $z_1=x_1, z_2=x_2, z_3=k_{3,1} \cdot x_1 + x_3$
- Shear along x_3 also has two reference directions, x_1 and x_2

Figure 20 · Shear Along x_3 , Relative to x_1 , in 3D

- The cube tilts along x_3 , with the amount set by the x_1 coordinate; volume is preserved



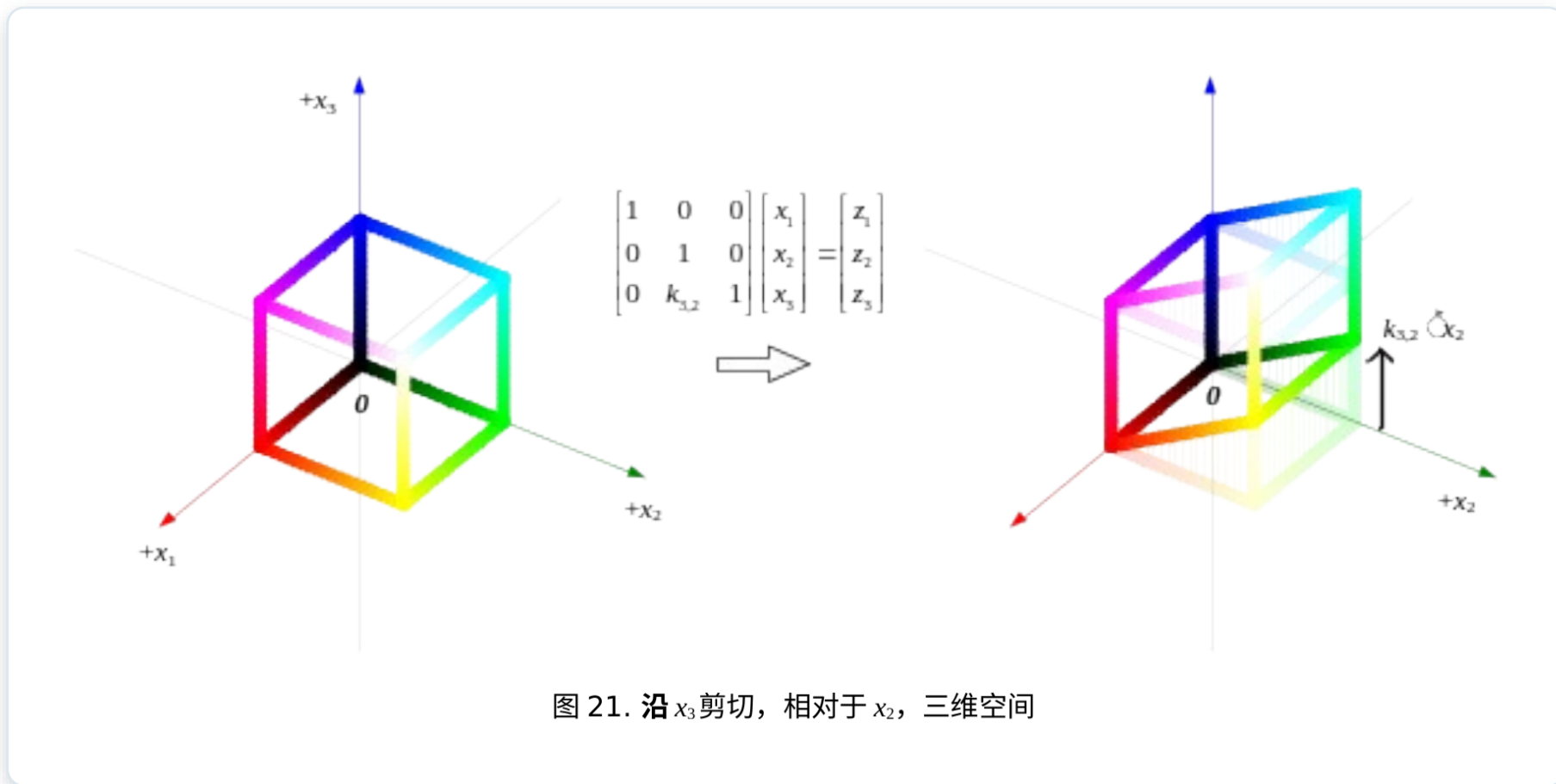
The shear in x_3 varies with x_2

$$[z_1; z_2; z_3] = [1, 0, 0; 0, 1, 0; 0, k_{3,2}, 1] @ [x_1; x_2; x_3]$$

- Expanded: $z_1=x_1, z_2=x_2, z_3=k_{3,2} \cdot x_2 + x_3$
- Try multiplying the two matrices from Figures 20 and 21, and explain the resulting geometric operation

Figure 21 · Shear Along x_3 , Relative to x_2 , in 3D

- The cube tilts along x_3 , with the amount set by the x_2 coordinate; volume is preserved



SUMMARY

What Did We Learn from Shear as Matrix Multiplication?

A shear matrix is upper or lower triangular with determinant 1 — area/volume is preserved

The shear angle ϑ unifies horizontal and vertical shear through $\cot(\vartheta)$

Shear matrices are invertible — the inverse just negates the shear factor k

Same-axis composition commutes; different-axis composition generally does not

A nonzero diagonal matrix can be written “scale@shear” or “shear@scale” — the two are transposes

Key Terms in This Section

Shear 剪切 —— A transform that keeps one direction's coordinate fixed while shifting the other in proportion

Vertical Shear 竖直剪切 —— x_1 stays fixed; x_2 shifts in proportion to x_1

Shear Angle 剪切角 —— ϑ , satisfying $k = \cot(\vartheta)$; also controls direction and strength

Lower Triangular Matrix 下三角矩阵 —— all entries above the main diagonal are zero

Scaling Matrix 缩放矩阵 —— S , combines with shear matrix K to decompose any nonzero-diagonal triangular matrix

Horizontal Shear 水平剪切 —— x_2 stays fixed; x_1 shifts in proportion to x_2

Shear Factor 剪切因子 —— k , a ratio controlling direction and strength, not a translation

Upper Triangular Matrix 上三角矩阵 —— all entries below the main diagonal are zero

Commutative Law 交换律 —— holds for same-axis shear composition, generally not for different-axis

3D Shear 三维剪切 —— a shear along one axis can depend on either of the other two, six basic forms in all

FURTHER STUDY

Extended Reading

Review · 8.2 Scaling

The scaling matrix S underlying this section's scale-shear decomposition

Review · 8.3 Rotation

Another common 2×2 planar geometric transform, alongside shear

Further Reading · 13.2 Cholesky Decomposition

This section's scale-shear decomposition deepens into LDL and Cholesky decomposition

Open-Source Repository

Code and figures for the full book: <https://github.com/Visualize-ML>

Q1 – Q9

Q1 Decide whether each of $[1,-1;0,1]$, $[1,-1;0,2]$, $[1,0;1,1]$, $[2,0;1,1]$, $[2,1;0,0]$ is a planar shear matrix, and explain why.

Q3 Explain why a planar shear transform never changes a figure's area.

Q5 Compute $[1,1;0,1]$ and $[1,0;1,1]$ each raised to the 5th power (via repeated matrix multiplication), and explain the results geometrically.

Q7 Modify LA_08_02_01.ipynb to visualize the planar shears in Figures 1–9.

Q9 Explain the geometric operation in 4D represented by $[1,0,k,0; 0,1,0,0; 0,0,1,0; 0,0,0,1]$.

Q2 Given points $[(1,0),(0,1),(-1,0),(0,-1),(0,0)]$, write Python code applying the shear matrix $[1,0;1,1]$ and print the new coordinates.

Q4 Apply a horizontal shear with factor $k=2$ to the point $(3,4)$; use matrix multiplication to find the transformed coordinates.

Q6 Write the diagonal matrices $[1,-1;0,2]$ and $[2,0;1,1]$ each as “scale@shear” and as “shear@scale”.

Q8 Modify LA_08_03_01.ipynb to visualize the 3D shears in Figures 14–21.

Q10 – Q17

Q10 Use numpy to verify that every shear matrix has determinant equal to 1.

Q12 Use matplotlib to visualize a unit square's continuous deformation as the shear factor k ranges from -2 to 2.

Q14 Verify in code that two same-axis shears with different factors compose into a shear whose factor is their sum.

Q16 Write a Python function `is_shear_matrix(M)` that tests whether a given 2×2 matrix is a shear matrix.

Q11 Write a Python function `shear_matrix(k, axis)` that returns the 2×2 shear matrix along the given axis.

Q13 Explore $k = \cot(\theta)$: plot k against θ in Python, and observe the limiting behavior as θ approaches 0° and 90° .

Q15 Verify in code that two different-axis shears do not commute, and compute the difference between the two orderings.

Q17 Use `numpy.linalg.inv` to invert several shear matrices, and confirm the results are still shear matrices of the same type.

Q18 – Q25

Q18 Explore: given a 2×2 matrix with nonzero diagonal, write Python code that automatically finds both its “scale@shear” and “shear@scale” decompositions.

Q20 Verify in code that a 3D shear matrix's determinant is always 1, so volume is preserved under the transform.

Q22 Use Python to generate a series of static plots approximating the continuous 3D shear as $k_{1,2}$ varies from 0 to 2.

Q24 Comprehensive: given a 4×4 matrix with three off-diagonal shear factors and identity elsewhere, use code to identify the composite geometric operation.

Q19 Use Python to compare shearing along x_1 then x_2 in 3D with the reverse order — is the final matrix the same?

Q21 Explore generalizing 2D shear to n dimensions with a single off-diagonal shear factor k , and verify in code that the determinant stays 1.

Q23 Explore the link between LDL decomposition and shear matrices: run an LDL decomposition on a symmetric positive definite matrix in Python, and observe the lower-triangular shear factors.

Q25 Preview: compare shear and orthogonal projection (Section 8.5) in terms of whether they preserve area/volume and whether they are invertible.

10 Prompts to Explore with AI

- 1 Explain why the shear matrix $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$ always has determinant 1, and what that means geometrically.
- 2 Use Python to demonstrate how a square continuously deforms as the shear factor k goes from negative to positive.
- 3 Explain the relation $k = \cot(\theta)$ between shear factor and shear angle, and what shear state corresponds to θ near 0° and 90° .
- 4 Prove that two same-axis shear matrices commute under multiplication.
- 5 Explain why two different-axis shear matrices generally do not commute, illustrated with a concrete numeric example.

10 Prompts to Explore with AI

6 Explain why the “scale@shear” and “shear@scale” decompositions of a matrix are transposes of each other.

7 Explain the possible connection between shear transforms and the Cholesky and LDL decompositions.

8 Use Python to visualize the combined 3D shear along x_1 relative to both x_2 and x_3 .

9 Explain why shear preserves area or volume, yet changes angles and shape.

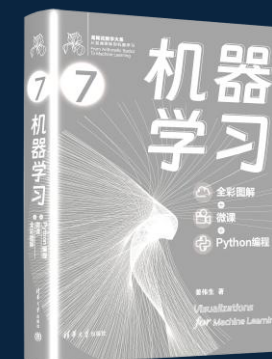
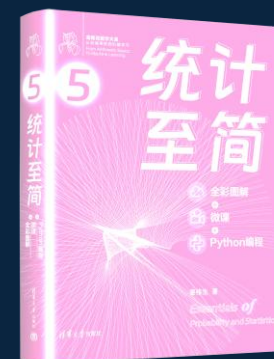
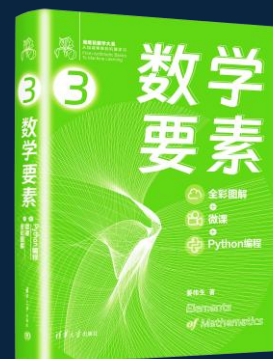
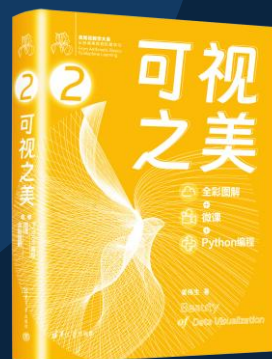
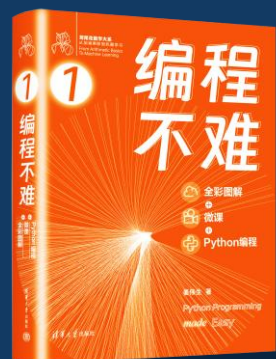
10 Explain what applications shear transforms might have in computer graphics, image processing, or generating italic fonts.

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

